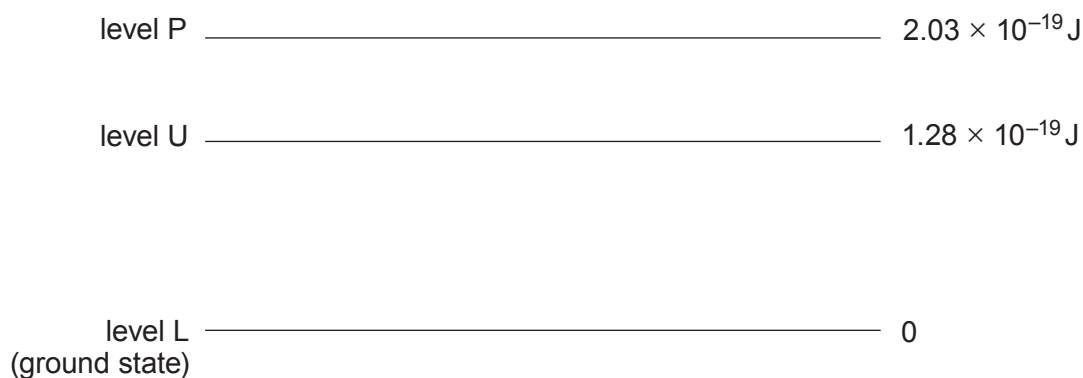


8. (a) A simplified energy level diagram for a 3-level laser system is given.



- (i) Calculate the wavelength of photons which can take part in transitions between levels U and L. [2]

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- (ii) Describe the processes of *absorption* and *stimulated emission* involving levels U and L. [4]

absorption:

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stimulated emission:

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- (iii) If the relative populations of levels P, U and L are 0.2%, 54.0% and 45.8%, state with a reason whether absorption or stimulated emission is the more likely event for a photon of the wavelength calculated in (a)(i). [1]

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- (iv) The laser is pumped by promoting electrons from the ground state to level P. Explain why electrons must spend only a very short time in level P, but a relatively long time in level U. [2]

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- (b) It has been suggested that energy should be carried across the country as high power light beams from lasers. Discuss whether or not this might be a good alternative to using electrical power lines. [4]

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END OF PAPER

